

M.SC. II YEAR BIOCHEMISTRY • QUICK RECALL REVISION

PAGE 1 • CHEMICAL FOUNDATIONS: BONDS, INTERACTIONS, EQUILIBRIA, KINETICS & ENERGETICS

M.SC. PART II

UNIT 1: FOUNDATIONS

POSTGRADUATE LEVEL

1. ATOMS & BONDS Covalent & Ionic Framework $\Delta\chi$ dictates bond polarity	→	2. MOLECULAR FORCES H-bonds, vdW, Hydrophobic Solvent entropy-driven folding	→	3. PH & BUFFERS Henderson-Hasselbalch Open HCO_3^- & cellular buffers	→	4. KINETICS 0, 1 st , 2 nd Order rates Concentration dependence	→	5. THERMODYNAMICS $\Delta G = \Delta H - T\Delta S$ Spontaneity $\neq$ Reaction Speed	→	6. COLLIGATIVE Particle number dependent Osmolality vs Osmolarity
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CHEMICAL BONDS IN BIOMOLECULES: COMPARATIVE MATRIX

ΔX & BOND STRENGTHS

BOND TYPE	MECHANISM & ΔX	ENERGY (KJ/MOL)	BIOCHEMICAL ROLES & SPATIAL CONSTRAINTS
Covalent (Non-Polar)	Equal electron sharing $\Delta\chi < 0.4$	350 – 460	C–C (348), C–H (414). Primary backbone; free rotation around σ -bonds unless hindered.
Covalent (Polar)	Unequal sharing; dipole $\Delta\chi = 0.4 - 1.7$	300 – 460	O–H (463), N–H (393), C=O (745). Permanent dipoles; reactive electrophilic/nucleophilic centers.
Ionic / Salt Bridge	Coulombic charge attraction $\Delta\chi > 1.7$; $F \propto q_1q_2/(\epsilon r^2)$	40 – 200 (gas) 12 – 20 (in H_2O)	$\text{Asp}^- / \text{Glu}^- \dots \text{Lys}^+ / \text{Arg}^+$. Heavily screened by water dielectric ($\epsilon \approx 80$) and free counterions.
Resonance Hybrid	Delocalized π -electrons across multiple p-orbitals	+80 – 120 extra stability	Peptide bond (C–N 1.32 Å), Benzene ring, Guanidinium (Arg). Planar, rigid; suppresses rotation.

ELECTRONEGATIVITY (χ) TREND & POLAR BOND SCHEMATIC:
 $F (4.0) > O (3.5) > N (3.0) \approx Cl (3.0) > C (2.55) \approx S (2.58) > H (2.20) \approx P (2.19)$
 $\delta+ H \longrightarrow O \delta- \quad \delta+ C \longrightarrow O \delta- \quad \delta+ C \longrightarrow N \delta- \quad \mu = q \times r \text{ (Dipole Vector)}$

BIOLOGICAL NON-COVALENT FORCES

DYNAMIC & REVERSIBLE

Hierarchy: Covalent bonds = primary immutable skeletal framework; Non-covalent forces = 3D folding, macromolecular recognition & flexible transition states.

- Hydrogen Bonds (12–30 kJ/mol):** Collinear D–H...A; distance 2.7–3.1 Å. Highly directional; stabilizes α -helices, β -sheets, DNA base pairing.
- Electrostatic / Salt Bridges (12–20 kJ/mol):** Between opposite full charges; sensitive to pH & ionic strength; stabilize buried interior binding pockets.
- van der Waals Interactions (0.4–4 kJ/mol):** Induced transient dipoles; potential $\propto 1/r^6$. Individually weak, massive cumulative stabilizing effect in packed cores.
- Hydrophobic Effect (Entropic Driver):** NOT an attractive force! Aggregation of nonpolar residues releases ordered water cages into bulk ($\Delta S_{\text{water}} > 0$).
- Steric Repulsion ($\propto 1/r^{12}$):** Pauli exclusion prevents atomic interpenetration; confines polypeptide backbone to allowed Ramachandran angles.

⚠ DON'T CONFUSE: RESONANCE HYBRID VS DYNAMIC SWITCHING
A resonance hybrid is a **single real quantum intermediate state** of minimum energy, **NOT** rapid oscillation or dynamic equilibrium between two canonical structures!

Δ ACID-BASE CHEMISTRY, HENDERSON-HASSELBALCH & PHYSIOLOGICAL BUFFER SYSTEMS

BUFFER RANGE: $\text{pH} = \text{pK}_a \pm 1.0$

KEY EQUATIONS & THREE STATES
 $\text{pH} = -\log[\text{H}^+] \quad | \quad \text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$

- $\text{pH} = \text{pK}_a$:** $[\text{A}^-] = [\text{HA}]$ (50% ionized). Titration midpoint; **Maximal Buffer Capacity (β)**.
- $\text{pH} > \text{pK}_a$:** $[\text{A}^-] > [\text{HA}]$ (Deprotonated/conjugate base dominates). At $\text{pK}_a + 1$, ratio is 10:1 (90.9% deprotonated).
- $\text{pH} < \text{pK}_a$:** $[\text{HA}] > [\text{A}^-]$ (Weak acid dominates). At $\text{pK}_a - 1$, ratio is 1:10 (90.9% protonated).
- Buffer Capacity (β):** $\beta = \Delta B / \Delta \text{pH} = 2.303 \times C_{\text{tot}} \times ([\text{HA}][\text{A}^-]/C_{\text{tot}}^2)$. Proportional to total buffer concentration C_{tot} .

PHYSIOLOGICAL BUFFER SYSTEMS

- 1. Bicarbonate Buffer (ECF/Plasma, $\text{pH} 7.4$):**
 $\text{CO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$ ($\text{pK}_a' = 6.1$)
Open system: $[\text{CO}_2]$ expired by lungs; $[\text{HCO}_3^-]$ reabsorbed by kidneys.
- 2. Phosphate Buffer (Cytosol, $\text{pK}_a = 6.86$):**
 $\text{H}_2\text{PO}_4^- \rightleftharpoons \text{H}^+ + \text{HPO}_4^{2-}$. Crucial in intracellular fluid & renal tubular lumen.
- 3. Protein & Hemoglobin Buffers:**
Histidine imidazole ($\text{pK}_a \approx 6.0\text{--}7.0$). DeoxyHb binds H^+ more avidly than OxyHb (Bohr effect, isohydric transport).

TITRATION CURVE & BUFFERING WINDOW

∞ CHEMICAL KINETICS MATRIX

ORDER VS RATE

ORDER	DIFFERENTIAL RATE	INTEGRATED EQUATION	HALF-LIFE ($T_{1/2}$)	UNITS (K)
0	Rate = k	$[\text{A}]_t = [\text{A}]_0 - kt$	$[\text{A}]_0 / (2k)$	$\text{M}\cdot\text{s}^{-1}$
1 st	Rate = k[A]	$\ln[\text{A}]_t = \ln[\text{A}]_0 - kt$	$0.693 / k$	s^{-1}
2 nd	Rate = k[A] ²	$1/[\text{A}]_t = 1/[\text{A}]_0 + kt$	$1 / (k[\text{A}]_0)$	$\text{M}^{-1}\cdot\text{s}^{-1}$

Zero-order: Rate independent of [A]; enzyme saturated ($[\text{S}] \gg K_m$).
1st-order: $t_{1/2}$ is strictly independent of $[\text{A}]_0$!
2nd-order: $t_{1/2}$ is inversely proportional to initial $[\text{A}]_0$.

ΔG BIOCHEMICAL THERMODYNAMICS

ΔG = ΔH – TΔS

$\Delta G = \Delta H - T\Delta S \quad | \quad \Delta G' = -RT \ln K'_{\text{eq}}$

- ΔG < 0 (Exergonic):** Thermodynamically spontaneous; releases free energy. $K'_{\text{eq}} > 1$.
- ΔG > 0 (Endergonic):** Non-spontaneous; driven by ATP coupling ($\Delta G'' \approx -30.5 \text{ kJ/mol}$). $K'_{\text{eq}} < 1$.
- ΔG = 0 (Equilibrium):** $\text{Rate}_{\text{forward}} = \text{Rate}_{\text{reverse}}$; zero net flux.
- ΔG = ΔG'' + RT ln Q:** Actual cellular ΔG depends on instantaneous mass-action ratio Q!

⚠ EXAM TRAP: SPONTANEOUS ≠ FAST!
Thermodynamics (ΔG) dictates *feasibility & equilibrium position*. Kinetics (activation energy E_a / $\Delta G^\ddagger$) dictates *velocity*! Sucrose oxidation has $\Delta G'' \approx -5690 \text{ kJ/mol}$ but $t_{1/2} > 1000 \text{ yrs}$ without invertase.

Π COLLIGATIVE & OSMOTIC PROPERTIES

PARTICLE NUMBER

Colligative Principle: Properties depend strictly on the *number of solute particles* in solution, not their size, charge, or chemical structure!

- Osmotic Pressure: $\Pi = i\text{CRT}$ (van 't Hoff equation)
- Freezing Pt Depression: $\Delta T_f = i K_f m$ (Cryoscopy)
- Boiling Pt Elevation: $\Delta T_b = i K_b m \quad | \quad \Delta P = X_B P^\circ$

⚠ DON'T CONFUSE: OSMOLARITY VS OSMOLALITY

- Osmolarity:** Osmoles / Liter solution (temp-dependent; volume expands with T).
- Osmolality:** Osmoles / kg solvent (temp-independent; clinical standard: 285–295 mOsm/kg).

**van 't Hoff factor i: NaCl → i ≈ 1.85 (dissociates into 2 ions); Glucose → i = 1.0 (non-electrolyte).*

⚡ PAGE 1 MUST REMEMBER HIGH-YIELD FORMULAE & CONCEPTS

1. $\text{pH} = \text{pK}_a \pm 1$ (Buffer window; at $\text{pH} = \text{pK}_a$, $[\text{HA}] = [\text{A}^-]$)

2. $\Delta G'' = -RT \ln K'_{\text{eq}}$ (–5.7 kJ/mol per 10-fold K'_{eq} shift)

3. 1st-order $t_{1/2} = 0.693/k$ (conc-independent)

4. Hydrophobic effect is driven by water entropy ($\Delta S_{\text{water}} > 0$)

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PAGE 2 • CARBOHYDRATES & LIPIDS: STRUCTURE, CLASSIFICATION, CONJUGATES & MEMBRANES

M.SC. PART II

UNIT 2: BIOMOLECULES

STRUCTURAL MATRIX

1. MONOSACCHARIDES Aldose/Ketose, D/L C5 Epimers (Gal C4, Man C2) & α/β	→	2. OLIGO & POLY Maltose, Lactose, Sucrose Storage vs Structural polymers	→	3. GLYCOCONJUGATES Glycoprotein vs Proteoglycan GAGs, ECM, Cell Recognition	→	4. FATTY ACIDS Sat vs Cis-unsaturated kinks Melting temp (T _m) determinants	→	5. NEUTRAL TAGS Glycerol + 3 FAs Pure storage (6× energy vs Glycogen)	→	6. MEMBRANE LIPIDS Phospho- & Sphingolipids Sterols & Lipid Raft Dynamics
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◇ CARBOHYDRATES CLASSIFICATION, STEREOCHEMISTRY & POLYSACCHARIDES

STEREO RULES & LINKAGES

Stereochemical Rules & Isomerism:

- **D- vs L-Series:** Assigned by configuration of chiral carbon *furthest from carbonyl* (C5 in hexoses). Naturally occurring vertebrate sugars are predominantly **D-isomers**.
- **Epimers:** Diastereomers differing at only one stereocenter: **D-Glucose ⇌ D-Galactose at C-4** ; **D-Glucose ⇌ D-Mannose at C-2**.
- **Anomers (α vs β):** Ring closure forms hemiacetal/hemiketal with new stereocenter at C1. In D-sugars: **α-anomer** has -OH opposite to -CH₂OH (down in Haworth); **β-anomer** has -OH cis to -CH₂OH (up in Haworth, equatorial in β-D-Glucopyranose → sterically favored, 64% at equilibrium).
- **Mutarotation:** Spontaneous interconversion between α and β anomers via open-chain intermediate until equilibrium optical rotation is reached.

Disaccharides & Glycosidic Bonds:

- **Maltose:** α-D-Glcp-(1→4)-D-Glcp • **Reducing** (free hemiacetal at C1).
- **Lactose:** β-D-Galp-(1→4)-D-Glcp • **Reducing** (cleaved by β-galactosidase/lactase).
- **Sucrose:** α-D-Glcp-(1⇌2)-β-D-Fruf • **NON-REDUCING!** (both anomeric carbons participate in glycosidic linkage; no free hemiacetal).
- **Trehalose:** α-D-Glcp-(1⇌1)-α-D-Glcp • **NON-REDUCING** (insect hemolymph, cryoprotectant).

POLYSACCHARIDE	MONOMER & LINKAGE	BRANCHING PATTERN	3D CONFORMATION & BIOLOGICAL FUNCTION
Starch (Amylose)	D-Glucose α(1→4) linear	None (unbranched)	Left-handed open helix; traps iodine (I ₃ ⁻) giving intense blue color. Plant storage.
Starch (Amylopectin)	D-Glucose α(1→4) main chain	α(1→6) branch points every 24 – 30 residues	Large branched clusters; plant energy reservoir. Hydrolyzed by salivary & pancreatic α-amylase.
Glycogen	D-Glucose α(1→4) main chain	α(1→6) branch points every 8 – 12 residues	Highly compact globular polymer in liver & muscle. Multi-branched structure provides thousands of non-reducing ends for rapid simultaneous phosphorylase.
Cellulose	D-Glucose β(1→4) unbranched	None (linear ribbons)	Extended flat sheets stabilized by extensive intra- and inter-chain H-bonds. Insoluble, high tensile strength; impervious to human amylase (requires cellulase).
Chitin	N-Acetylglucosamine (GlcNAc) β(1→4) linear	None	Hard, water-insoluble structural component of arthropod exoskeletons and fungal cell walls.

≡ GLYCOCONJUGATES: GLYCOPROTEINS, PROTEOGLYCAN & GLYCOLIPIDS

ECM, SIGNALING & GLYCOLALYX

1. GLYCOPROTEINS

- **Mass ratio:** Protein > Carbohydrate (typically 1 – 60% carbohydrate by mass).
- **Glycan structure:** Short, branched oligosaccharides, non-repeating, often sialylated.
- **Linkages:**
 - **N-linked:** β-glycosidic to amide N of **Asn** in consensus sequence: Asn-X-Ser/Thr (X ≠ Pro).
 - **O-linked:** α-glycosidic to -OH of **Ser, Thr,** or hydroxylysine.
- **Roles:** Antibodies (IgG), cell receptors, blood group antigens, mucins, viral hemagglutinin targets.

2. PROTEOGLYCAN & GAGs

- **Mass ratio:** Carbohydrate >> Protein (typically >90% **GAG** by mass).
- **Glycosaminoglycans (GAGs):** Long, linear, unbranched repeating disaccharides composed of [Uronic acid (D-glucuronic/L-iduronic) + Amino sugar (GlcNAc/GalNAc)].
- **High negative charge:** Heavy sulfation (–SO₃⁻) and carboxylates attract Na⁺ and massive hydration spheres, creating high turgor pressure.
- **Types:** Chondroitin sulfate (cartilage), Heparan sulfate (basement membrane), Keratan sulfate (cornea), Hyaluronic acid (non-sulfated, unattached to core protein, synovial fluid lubricant).

3. GLYCOLIPIDS & FUNCTIONS

- **Structure:** Membrane sphingolipid or glycerol backbone with covalently linked polar carbohydrate head.
- **Gangliosides:** Complex oligosaccharide with ≥1 **Sialic acid (NANA)** residue. Net negative charge; enriched in neural synapses.
- **ABO Determinants:** Specific terminal sugar additions on glycolipids/glycoproteins: O antigen (H substance) + GalNAc = A antigen; + Gal = B antigen.
- **Receptor function:** Ganglioside GM1 is the direct mucosal receptor for cholera toxin.

Δ FATTY ACIDS & TRIACYLGLYCEROLS (NEUTRAL STORAGE LIPIDS)

SATURATED VS CIS-KINKS

Saturated Fatty Acids:

- No C=C double bonds. Fully extended zigzag conformation.
- Pack tightly via maximal van der Waals contacts.
- **High T_m:** solid at room temp (e.g. Palmitic 16:0, T_m=63°C; Stearic 18:0, T_m=69.6°C).

Unsaturated Fatty Acids (Cis):

- Natural double bonds are almost exclusively **cis**.
- Cis bond introduces rigid **~30° kink**, preventing tight packing.
- **Low T_m:** liquid oils at room temp (e.g. Oleic 18:1^{Δ9}, T_m=13.4°C; Linoleic 18:2^{Δ9,12}, T_m=-5°C).

Lipid Standard Notation: • **Delta (Δ) System:** Carboxyl carbon = C1. 18:1^{Δ9} = 18 carbons, 1 cis double bond at C9-C10. • **Omega (ω / n) System:** Methyl carbon = ω1. ω-3 = first double bond at C3 from methyl end (α-linolenic 18:3); ω-6 = linoleic acid 18:2.

Triacylglycerols (TAGs / Triglycerides): Glycerol esterified with 3 fatty acids.

- **Complete hydrophobicity:** Neutral, zero charge, nonpolar. Completely excluded from water.
- **Anhydrous storage:** Yields ~38 kJ/g (9 kcal/g) vs ~17 kJ/g for carbs. Glycogen binds 2 g H₂O per gram; therefore TAGs provide ~6× more metabolic energy per gram wet weight!

MEMBRANE BILAYER ARCHITECTURE & STEROLS

AMPHIPHATHIC CORE

- **Cholesterol (Fluidity Buffer):** Rigid steroid ring (4 fused rings) + polar 3β-OH. Functions as a **bidirectional fluidity buffer**: prevents excessive fluidity and leaks at high T; prevents acyl-chain crystallization and freezing at low T.
- **Lipid Rafts:** Microdomains enriched in Cholesterol + Sphingolipids + GPI-anchored proteins; serve as signaling platforms.

◇ MEMBRANE & STORAGE LIPIDS: STRUCTURAL MATRIX						BACKBONE & HEAD GROUP CHEMISTRY
LIPID CLASS	BACKBONE	HYDROPHOBIC TAIL	POLAR HEAD GROUP	NET CHARGE (PH 7.4)	PRIMARY BIOLOGICAL ROLE	
Triacylglycerol (TAG)	Glycerol	3 Fatty Acids (ester linkage)	None (hydrophobic)	0 (Neutral)	Dense anhydrous metabolic energy store (adipocytes).	
Glycerophospholipid (e.g. PC, PE)	Glycerol-3-P	2 Fatty Acids (C1 sat, C2 unsat)	Phosphate + Choline / Ethanolamine	0 (Zwitterion)	Major structural constituent of cellular membranes.	
Phosphatidylserine (PS)	Glycerol-3-P	2 Fatty Acids	Phosphate + Serine	-1	Inner leaflet marker; flips to outer leaflet during apoptosis.	
Cardiolipin	Diphosphatidyl-glycerol	4 Fatty Acyl chains	2 Phosphates + central glycerol	-2	Inner mitochondrial membrane; supports electron transport.	
Sphingomyelin	Sphingosine	1 Fatty Acid (amide bond to C2)	Phosphocholine / Ethanolamine	0 (Zwitterion)	Myelin sheath insulation in axons; lipid rafts.	
Cerebroside	Sphingosine	1 Fatty Acid (amide bond)	Single Sugar (Glucose/Galactose)	0 (Neutral)	Myelin lipid; neutral glycosphingolipid.	
Ganglioside (e.g. GM1)	Sphingosine	1 Fatty Acid (amide bond)	Complex oligosaccharide + ≥1 Sialic acid	-1 to -4	Neurotransmission, cell-surface recognition, toxin receptor.	

⚠ EXAM TRAP: TAG ≠ MEMBRANE LIPID!

TAGs are completely nonpolar, hydrophobic, and uncharged. They **CANNOT form bilayers** or liposomes. Membrane lipids (Glycerophospholipids & Sphingolipids) are **AMPHIPHATHIC**, possessing distinct hydrophilic heads and hydrophobic tails.

⚠ DON'T CONFUSE: GLYCOPROTEIN VS PROTEOGLYCAN VS GLYCOLIPID

- **Glycoprotein:** Protein > carb (short branched glycans; cell receptors/antibodies).
- **Proteoglycan:** Carb >> protein (>90% linear polyanionic GAGs; ECM cushion).
- **Glycolipid:** Sphingolipid/glycerol + sugar; anchored in outer bilayer leaflet.

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PAGE 3 • VITAMINS, HORMONES, AMINO ACID CLASSIFICATION & THE PEPTIDE BOND

M.SC. PART II

UNIT 3: COFACTORS & MONOMERS

HIGH-YIELD RECALL

1. FAT-SOLUBLE VITS
A, D, E, K Mechanisms
Deficiencies & toxicity risk

2. WATER-SOLUBLE
B/C
Precursors to Coenzymes
B1–B12 + C functional roles

3. VIT D SIGNALING
Nuclear VDR-RXR axis
Ca²⁺ homeostasis & CNS
notes

4. HORMONES
Peptide vs Steroid vs Tyr
Membrane vs Nuclear
receptors

5. AMINO ACIDS
5 Classes, Charges, pK_R
Essential (PVT TIM HALL)

6. PEPTIDE BOND
Planar, trans, 40% C=N
Zero rotation around ω

FAT-SOLUBLE VITAMINS (A, D, E, K): ACTIVE FORMS & DEFICIENCIES

LIPOPHILIC STORAGE

VIT	ACTIVE FORM(S)	BIOCHEMICAL ROLE	CLASSIC DEFICIENCY SIGNS
A	11- <i>cis</i> -Retinal Retinoic acid	Visual rhodopsin cycle in rods; nuclear RAR/RXR transcription factor regulating epithelial differentiation.	Nyctalopia (night blindness), Xerophthalmia, Bitot's spots, follicular keratosis.
D	1,25-(OH) ₂ D ₃ (Calcitriol)	Nuclear VDR receptor; stimulates intestinal Ca ²⁺ & P _i absorption via calbindin; bone mineralization.	Rickets (children: bowed legs, rachitic rosary); Osteomalacia (adults: demineralization, fractures).
E	α-Tocopherol	Membrane-resident lipid-soluble chain-breaking antioxidant; scavenges peroxy radicals, halts lipid peroxidation.	Hemolytic anemia (erythrocyte fragility), spinocerebellar ataxia, peripheral neuropathy.
K	Phylloquinone (K ₁) Menaquinone (K ₂)	Co-substrate for γ-glutamyl carboxylase (γ-carboxylation of Glu to Gla in Clotting Factors II, VII, IX, X & Protein C/S).	Hemorrhagic disease of newborn; prolonged Prothrombin Time (PT). Inhibited by Warfarin/Dicumarol.

Vitamin D Axis & Neuro Significance: 7-Dehydrocholesterol (skin+UVB) → D₃ → 25-(OH)D₃ (liver) → 1,25-(OH)₂D₃ (kidney 1α-hydroxylase, stimulated by PTH). Calcitriol binds nuclear **VDR** → heterodimerizes with **RXR** → binds **VDRE** to induce Calbindin-D_{28k} and TRPV6.

**Neuro note (cautious association):* VDR and 1α-hydroxylase are expressed in CNS neurons and glia; epidemiological data correlate deficiency with cognitive changes, though direct causal neuroprotective mechanisms remain under ongoing clinical evaluation.

WATER-SOLUBLE B-COMPLEX & VITAMIN C COENZYME MATRIX

DIRECT COENZYMES

B1 (Thiamine → TPP): Oxidative decarboxylation (Pyruvate DH, α-KGDH) & transketolase (HMP shunt).
Def: Beriberi (Wet: cardiac; Dry: neuropathy), Wernicke-Korsakoff.

B2 (Riboflavin → FAD, FMN): 2e⁻/2H⁺ redox carriers (Succinate DH, Acyl-CoA DH).
Def: Ariboflavinosis (cheilosis, angular stomatitis, magenta tongue).

B3 (Niacin → NAD⁺, NADP⁺): Hydride ion (H⁻) transfer. Catabolism (NAD⁺) vs biosynthesis (NADPH).
Def: Pellagra (4 Ds: Diarrhea, Dermatitis, Dementia, Death). From Trp (60:1).

B5 (Pantothenate → CoA-SH, ACP): Acyl group activation via high-energy thioester bond (ΔG^o ≈ -31.5 kJ/mol).
Def: Burning feet syndrome, paresthesias.

B6 (Pyridoxine → PLP): Transamination (AST, ALT: Schiff base), decarboxylation, glycogen phosphorylase.
Def: Microcytic sideroblastic anemia, seizures, neuropathy.

B7 (Biotin): Carrier of activated CO₂ in ATP-carboxylases (Pyruvate carboxylase, ACC).
Def: Dermatitis, alopecia; induced by raw egg white **Avidin**.

B9 (Folate → THF): 1-Carbon unit transfer (-CH₃, -CHO) in dTMP & purine synthesis.
Def: Megaloblastic anemia (hypersegmented neutrophils), Neural Tube Defects.

B12 (Cobalamin → Methyl/Ado-B12): Methionine synthase & Methylmalonyl-CoA mutase.
Def: Pernicious anemia (lack of IF), Subacute combined cord degeneration.

Vitamin C (Ascorbic Acid): Water-soluble antioxidant; co-substrate for Prolyl & Lysyl hydroxylases (keeps Fe²⁺ reduced for collagen triple-helix stability); enhances dietary Fe absorption. **Def:** Scurvy (bleeding gums, poor wound healing, corkscrew hair).

HORMONES: CHEMICAL CLASSIFICATION & RECEPTOR SIGNALING MECHANISMS

MEMBRANE VS NUCLEAR RECEPTORS

1. Peptide & Protein Hormones:

- **Examples:** Insulin, Glucagon, ACTH, PTH, TSH, Oxytocin.
- **Properties:** Hydrophilic, circulate free; bind **Cell-Surface Receptors** (GPCRs, RTKs).
- **Signaling:** Second messengers (cAMP, IP₃/DAG, Ca²⁺); protein phosphorylation cascades. *Rapid onset, short half-life (minutes).*

2. Steroid Hormones:

- **Examples:** Cortisol, Aldosterone, Estrogen, Testosterone.
- **Properties:** Hydrophobic (from cholesterol); bound to carrier proteins (CBG, SHBG, Albumin).
- **Signaling:** Diffuse across membrane → bind **Intracellular / Nuclear Receptors** → direct gene transcription. *Slow onset (hours), long half-life.*

3. Amino Acid-Derived (Tyrosine):

- **Catecholamines (Epi/NE):** Hydrophilic; stored in vesicles; bind membrane GPCRs (α & β adrenergic); fast fight-or-flight.
- **Thyroid Hormones (T₃, T₄):** Lipophilic (iodinated); transported on TBG; cross membrane via MCT8 → bind **Nuclear TR Receptors** → basal metabolic rate ↑.

AMINO ACIDS MASTER CLASSIFICATION (AT PH 7.0 - 7.4)

ESSENTIAL: PVT TIM HALL

CHEMICAL CLASS	AMINO ACIDS (3-LETTER & 1-LETTER)	SIDE-CHAIN Pk _R	PHYSICAL & CORE STRUCTURAL ROLES
Nonpolar Aliphatic	Gly (G), Ala (A), Val (V), Leu (L), Ile (I), Pro (P), Met (M)	—	Buried in hydrophobic interior core. Pro = cyclic rigid imino acid; Gly = achiral, highly flexible.
Aromatic	Phe (F), Tyr (Y), Trp (W)	Tyr: 10.1	Hydrophobic packing. Trp absorbs UV at 280 nm (ε = 5600 M ⁻¹ cm ⁻¹); Tyr phenolic -OH.
Polar Uncharged	Ser (S), Thr (T), Cys (C), Asn (N), Gln (Q)	Cys: 8.3	Surface H-bonding. Cys -SH oxidizes to covalent Disulfide (-S-S-) crosslinks.
Acidic (Negative)	Aspartate (Asp, D), Glutamate (Glu, E)	Asp: 3.9 Glu: 4.1	Deprotonated (-COO ⁻) at pH 7. Form ionic salt bridges with Lys/Arg; bind Ca ²⁺ /Mg ²⁺ .
Basic (Positive)	Lysine (Lys, K), Arginine (Arg, R), Histidine (His, H)	Lys: 10.5 Arg: 12.5 His: 6.0	Lys & Arg are fully protonated (+) at pH 7. His imidazole (pK_R ≈ 6.0) shuttles H ⁺ as general acid-base catalyst.

Essential Amino Acids (PVT TIM HALL): Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine. (Arg essential during growth).

THE PEPTIDE BOND: PLANARITY & RESONANCE

40% DOUBLE BOND

Condensation Reaction:



Resonance & Spatial Constraints:

- **40% Partial Double-Bond Character:** Delocalization of N lone pair onto carbonyl O:



- **Bond Length:** C-N is **1.32 Å** (intermediate between single 1.45 Å and double 1.25 Å).
- **Rigid Coplanarity:** The 6 atoms (C_α¹, C, O, N, H, C_α²) lie strictly in a single plane.
- **Trans Configuration:** >99.9% in trans geometry to prevent steric clash between sidechains (Proline has ~6% cis due to pyrrolidine ring).
- **Rotation Rules: Zero rotation around peptide ω bond!** Rotation is permitted ONLY around single bonds flanking C_α: ϕ (N-C_α) and ψ (C_α-C).

PAGE 3 MUST REMEMBER RULES & HIGH-YIELD COENZYME MAPPINGS

1. B6 (PLP) = Transamination & Schiff base
2. B12 (Cobalamin) = Met synthase & odd-chain mutase
3. Histidine (pK_R ≈ 6.0) = Physiological catalytic buffer
4. Peptide bond = Planar, rigid trans, 1.32 Å, no ω rotation

M.SC. II YEAR BIOCHEMISTRY • QUICK RECALL REVISION

PAGE 4 • PROTEIN STRUCTURE, CONFORMATION, RAMACHANDRAN PLOT, STABILITY & FOLDING

M.SC. PART II

UNIT 4: PROTEIN BIOPHYSICS

FINAL MASTER SYNTHESIS

FOUR STRUCTURAL LEVELS & SECONDARY STRUCTURE ELEMENTS

HIERARCHY & H-BONDING

Four Structural Levels:

- **1° (Primary):** Covalent linear sequence of amino acids linked by peptide bonds.
- **2° (Secondary):** Local spatial backbone folding (α-helix, β-sheet) via $-C=O...H-N-$ H-bonds.
- **3° (Tertiary):** Overall 3D spatial conformation of a single chain; hydrophobic core.
- **4° (Quaternary):** Spatial arrangement of ≥ 2 subunits (e.g. Hb $\alpha_2\beta_2$).

α-Helix (Right-Handed):

- **3.6 residues/turn;** pitch = **5.4 Å** (0.54 nm); rise = 1.5 Å/res.
- **H-bond pattern:** Intrachain $C=O(i) \dots H-N(i+4)$.
- **Helix Dipole:** (+) at N-term, (−) at C-term.
- **Breakers:** Proline (kink, no backbone NH) & Glycine (too flexible). Ala & Leu favor helix.

β-Pleated Sheets:

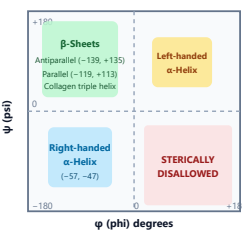
- Extended zigzag strands; inter-strand H-bonds.
- **Antiparallel:** Strands run opposite; H-bonds are linear (180°) → **maximally stable**; repeat = 7.0 Å.
- **Parallel:** Strands run same direction; H-bonds are skewed/angled → less stable; repeat = 6.5 Å.

β-Turns & Loops:

- **β-Turns (Reverse bends):** 4 residues reversing chain 180° ; $C=O(i) \dots H-N(i+3)$ bond.
- **Type I:** Proline at pos 2.
- **Type II:** Glycine strictly at pos 3.
- **Loops:** Surface flexible hydrophilic regions; ligand binding & active sites.

RAMACHANDRAN PLOT (Φ VS Ψ TORSION ANGLES)

STERIC ACCESSIBILITY



Ramachandran Coordinate Rules:

- **Top-Left (−Φ, +Ψ):** β-sheets (antiparallel & parallel) and polyproline II / collagen triple helix. Most extended conformation.
- **Bottom-Left (−Φ, −Ψ):** Right-handed α-helix ($\phi \approx -57^\circ$, $\psi \approx -47^\circ$) and 3_{10} -helix.
- **Top-Right (+Φ, +Ψ):** Left-handed α-helix ($\phi \approx +57^\circ$, $\psi \approx +47^\circ$, rare).
- **Bottom-Right (+Φ, −Ψ):** Disallowed for standard L-amino acids due to severe steric clash between side chain & main chain $C=O$.
- **Glycine:** Lacks C_β side chain; zero steric clash → allowed across **all 4 quadrants** (conformational flexibility).
- **Proline:** Pyrrolidine ring fixes ϕ rigidly at $-60^\circ \pm 15^\circ$; severely restricts allowed Ramachandran space.

STRUCTURAL HIERARCHY:

MOTIF ⊂ DOMAIN ⊂ PROTEIN

NESTED ARCHITECTURE

1. Motif (Supersecondary Structure):

Specific recurring spatial combination of secondary elements (e.g. β-α-β loop, Hairpin, Greek Key, Helix-Turn-Helix, Leucine zipper, Zinc finger). *Cannot fold independently or retain stability when isolated.*

2. Domain:

Compact, autonomously folding, structurally stable 3D functional unit of a polypeptide (typically 40–300 residues). *Retains its 3D native fold even if cleaved proteolytically from the protein* (e.g. SH2 domain, kinase domain, DNA-binding domain).

3. Protein Fold / Architecture:

Global topological arrangement of secondary structural elements forming a complete functional tertiary polypeptide or multi-subunit quaternary assembly.

FORCES STABILIZING 3D FOLD

CORE STABILITY

1. **Hydrophobic Effect:** Primary thermodynamic driver! Burial of nonpolar side chains releases water cages ($\Delta S_{\text{water}} > 0$, $\Delta G_{\text{fold}} < 0$).
2. **Disulfide Bonds (−S−S−):** Covalent crosslinks between Cys residues; stabilizes extracellular proteins (e.g. Insulin, RNase A).
3. **Hydrogen Bonds:** Backbone & sidechain H-bonds provide conformational specificity.
4. **Salt Bridges (Ion Pairs):** $\text{Asp}^-/\text{Glu}^- \dots \text{Lys}^+/\text{Arg}^+$. Sensitive to pH & ionic strength.
5. **van der Waals:** Tight core atom packing.

DENATURATION & ANFINSEN'S PROOF

1° DICTATES 3D FOLD

Denaturation: Unfolding of $2^\circ/3^\circ/4^\circ$ structure without breaking peptide bonds (Heat, 8M Urea, 6M GdnHCl, SDS, pH).

Anfinsen's Ribonuclease A Experiment:

Native RNase A (4 $-S-S-$) + 8M Urea + β-ME → 100% denatured/reduced (0% active).

• Dialysis (remove Urea FIRST, then air oxidize):

Spontaneously refolds to 100% active native state!

• Oxidize with Urea present: Scrambled inactive disulfides (1–2% active).

Conclusion: Primary sequence contains ALL information required to direct native 3D fold!

FOLDING FUNNEL & CHAPERONES

PREVENT AGGREGATION

Energy Funnel Landscape: High entropy/energy unfolded state → hydrophobic collapse → **Molten Globule intermediate** (native-like 2° structure, loose tertiary packing) → native minimum free energy ($\Delta G \approx -20$ to -60 kJ/mol).

Molecular Chaperones:

- **Hsp70 (DnaK):** Binds exposed hydrophobic patches on nascent chains to halt aggregation (ATP-cycle).
- **Chaperonins (GroEL/GroES):** Cylindrical double-ring cage isolating unfolded proteins for protected folding.

FINAL 60-SECOND MASTER RECALL COMPENDIUM

RAPID EXAM RECALL

- **pH = $pK_a + \log([A^-]/[HA])$:** At $pH=pK_a$, $[HA]=[A^-]$.
- **$\Delta G = \Delta H - T\Delta S$:** $\Delta G^\circ = -RT \ln K'_{eq} = -5.7$ kJ/mol for $10 \times K$.
- **Kinetics:** 1^{st} -order $t_{1/2} = 0.693/k$ (conc-independent!).
- **Osmotic Pressure:** $\Pi = iCRT$ (particle-number dependent).
- **Sucrose & Trehalose:** Non-reducing disaccharides!
- **TAG:** Neutral storage lipid (6× energy vs glycogen).
- **Peptide Bond:** Planar, trans, 40% double bond (1.32 Å).
- **α-Helix:** 3.6 res/turn, 5.4 Å pitch, $C=O(i) \dots H-N(i+4)$.
- **Ramachandran:** Gly = all 4 quadrants; Pro = restricted ($\sim 60^\circ$).
- **Anfinsen:** 1° sequence dictates native 3D fold.

HIGH-YIELD "DON'T CONFUSE" CRITICAL DISTINCTIONS

EXAM TRAPS

- **pH vs pK_a :** pH is solution acidity; pK_a is intrinsic acid dissociation constant.
- **ΔG vs Rate (k):** ΔG dictates feasibility/spontaneity; E_a & k dictate velocity!
- **Osmolarity vs Osmolality:** Osm/L (temp-dependent) vs Osm/kg (temp-independent).
- **Glycogen vs Cellulose:** Glycogen = $\alpha(1 \rightarrow 4) + \alpha(1 \rightarrow 6)$ storage; Cellulose = $\beta(1 \rightarrow 4)$ rigid fibril.
- **Motif vs Domain:** Motif = non-autonomous sub-structure; Domain = autonomous folding unit.
- **Denaturation vs Proteolysis:** Denaturation unfolds $2^\circ/3^\circ$ (peptide bonds INTACT); Proteolysis cleaves covalent peptide bonds!